Marien Chenaud

PhD Student in Applied Mathematics and Deep Learning

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`https://marien-chenaud.github.io/mypage/`

Formation

2022–2026 **PhD student**, *CentraleSupélec, Paris Saclay University*, Development of Digital twins for mechanical processes. I focus on demonstrating and implementing synergies between traditional numerical solvers and Deep Learning models to solve partial differential equations in complex, industrial domains. I implement my solutions in Python/Pytorch, and in Fortran. Related fields: Partial differential equations, Physics-informed neural networks, Finite Element method, Uncertainty Quantification. The PhD is part of an industrial chair with Transvalor SA.

2018–2022 **Engineering Degree**, *CentraleSupélec*, Mathematics and Data Science track.
Statistics, Machine Learning and Deep Learning, Functional Analysis, Physics, Computer Science

2020–2021 **Master's Degree in Mathematics of Randomness (Gap Year)**, *Paris-Saclay University*, Statistics and Machine Learning track
Stochastic Calculus, High-Dimensional Statistical Learning, Machine Learning Project

2020 **NTNU University**, *Trondheim, Norway*, Semester abroad (Erasmus program)
Optimisation, probabilistic and statistical modelisation, Climate modelisation, Philosophy.

2018–2019 **Bachelor's Degree in Fundamental and Applied Mathematics**, *Paris-Saclay University*

2016–2018 **Preparatory Classes for the Grandes Écoles**, *Lycée Thiers*, Marseille, Mathematics, Physics

2016 **French Scientific Baccalaureate, European Section, with Highest Honors**, Aix en Provence

Professional Experience and Projects

2021–2022 **Research project in computer vision**, *Study of Deep Learning models for pulmonary embolism detection*

2021 **Research Internship in Machine Learning (5 months)**, *French Alternative Energies and Atomic Energy Commission (CEA Saclay), Gap Year and End of Master's Degree*, Study of the coupling of Machine Learning and numerical simulations for fluid mechanics.

2020–2021 **Research Internship in Numerical Analysis (5 months)**, *CentraleSupélec Laboratory*, Study of partial differential equations for cancer tumor detection, numerical simulation using the finite element method, inverse problem.

2020 **Data Analysis Consulting Project**, Statistical study of a metro wear component for the Paris public transportation company, conducted as part of a student consulting contract at CentraleSupélec

Languages

French Native language

English Fluent (C1+)

Spanish Basic (B1)

Programming languages and softwares

Programming languages: Python (PyTorch, Numpy, Tensorflow, JAX), R, Fortran, FreeFem++

Softwares and tools: LaTeX, Git, Inkscape

Interests

Music Violin

Sports Handball, skiing, hiking

Others Chess, photography...

Publications

2026 **M. Chenaud, J. Alves, F. Magoulès**, *How to learn Physics - A practical review of Scientific Machine Learning*, Submitted to Archives of Computational Methods in Engineering.

2026 **M. Chenaud, J. Alves, F. Magoulès**, *A Numerical Approach to Physics-Informed Learning for Non-Analytic Problems*, Submitted to the Journal of Computational Physics.

2026 **M. Chenaud, J. Alves, F. Magoulès**, *Enhancing information flow in Graph Neural Networks through Geometry-Preserving Hierarchical Connectivity*, Submitted to Computers & Structures.

2026 **N. Anwer, M. Chenaud, D. Elizondo**, *Certificates for Complex-Compatible Learned Cochain Laplacians*, ICML 2026 (main track)

2025 **M. Chenaud, J. Alves, F. Magoulès**, *Enhancing information flow in graph neural networks for scientific machine learning*, The Seventh International Conference on Artificial Intelligence, Soft Computing, Machine Learning and Optimization in Engineering.

2024 **M. Chenaud, F. Magoulès, J. Alves**, *Physics-Informed Graph-Mesh Networks for PDEs: A hybrid approach for complex problems*, Advances in Engineering Software, 2024, 197, pp.103758

2023 **M. Chenaud, F. Magoulès, J. Alves**, *Physics-Informed Graph Convolutional Networks: Towards a generalized framework for complex geometries*, The Sixth International Conference on Soft Computing, ML and Optimization in Civil, Structural and Environmental Engineering.

Academic activities

Conferences and Seminars

2025 **Transvalor Tech days**, *Plenary talk*, Towards integrated Machine Learning in the engineering design cycle, Bergamo, Italy

2025 **The Seventh International Conference on Artificial Intelligence, Soft Computing, Machine Learning and Optimization in Engineering**, *Oral presentation*, Enhancing information flow in graph neural networks for scientific machine learning, Cagliari, Sardinia, Italy

2024 **Transvalor International Simulation Days**, *Oral Presentation*, Towards Digital Twins, Cannes, France

2023 **The Sixth International Conference on Soft Computing, Machine Learning and Optimization in Civil, Structural and Environmental Engineering**, *Oral presentation*, Physics-Informed graph Convolutional Networks, Pécs, Hungary.

2021 **TrioCFD Seminar**, *Poster Presentation*, Integration of Machine learning models for Computational Fluid Dynamics., French Alternative Energies and Atomic Energy Commission, France

Teaching

2023–2025 **Scientific Computing**, *CentraleSupélec*, Project Supervisor for Fourth-Year Engineering Students, Supervised theoretical and numerical investigations of physical models for noise-canceling walls.

2025 **Machine Learning**, *CentraleSupélec*, Practice sessions for bachelor students.

2023–2024 **Linear Algebra**, *CentraleSupélec*, Lectures, tutorials and examination for bachelor students. Assistance in the creation of the course curriculum.

2019 **Teaching Assistant in Mathematics (1 semester)**, *CentraleSupélec*, Tutorial and support sessions in Mathematics for third-year Engineering students.

2018–2019 **Private Mathematics Tutoring**, *Final Year of High School (Science Track)*

Academic Service

2026 **XXII Jacques-Louis Lions Hispano-French School on Numerical Simulation in Physics and Engineering**, *Support Committee*, Paris, France

2025 **Kinetic and Turbulence Equations Conference**, *Support to the organization*, CentraleSupélec - Collège de France, France